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Chapter 1

Introduction: Mott Transitions and Kondo Breakdown

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness is also responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

1.1 Metal-insulator transition arising from strong correlation

Initial attempts at explaining metal-insulator transitions was based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled

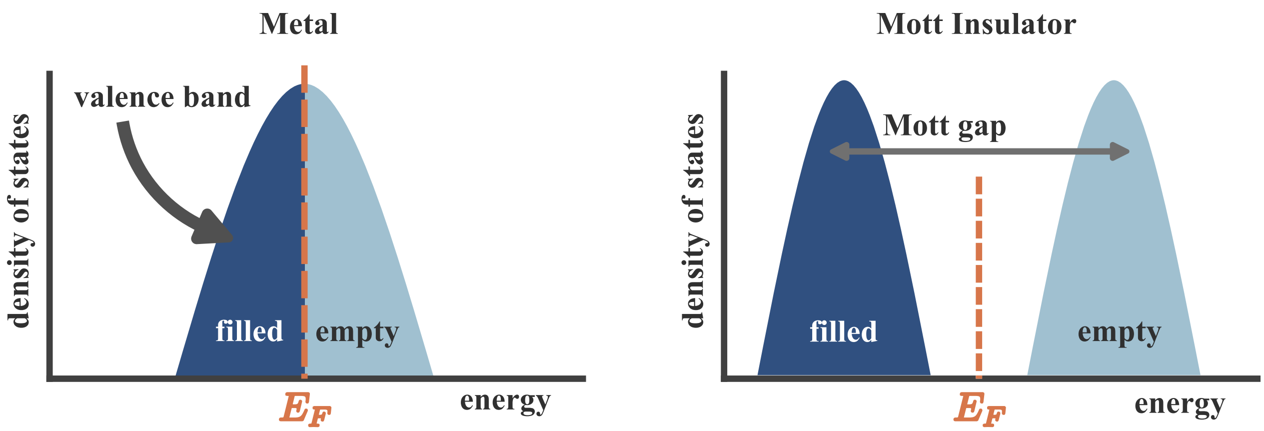

Figure 1.1

bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to an long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high-T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N \ . \tag{1.1}$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve $d-$electrons that are five-fold degenerate, and these orbitals can hybridisation with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard. Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$. We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probablity t. As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t, the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

- Brinkman-Rice vs Mott scenario

- Mott's holon-doublon binding-unbinding picture

- Greens function and self-energy properties

- spectral weight transfer

- Luttinger surfaces

1.3 Preliminary Insights Into Mottness

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1.8.1 Introduction to entanglement holography

1.8.2 Open questions

1.9 <title about an introduction to our holography work>

The present millennium has seen a rapid surge in the use of quantum entanglement towards the particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [27–30]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [23, 31, 32].

More pertinent to the present work is the entanglement of non-interacting relativistic fermionic systems. Such phases are known to emerge in several condensed matter systems [33], appearing at the surfaces of three-dimensional topological insulators [34–37], quasi-2D organic materials [38–40] and in artificial quantum simulators such as cold atom gases, molecular and microwave crystals [41–47]. Massless critical systems (or conformal field theories (CFTs)) differ from gapped systems due to the fact that their entanglement satisfies a modified area law; in one spatial dimension, the entanglement entropy (S) of a single subsystem of length l scales as $S \sim \frac{c}{3} \ln l$ [48–52], where c is the conformal central charge of the CFT. This violation arises from the non-local nature of the gapless quantum fluctuations that lie proximate to the $d-1$-dimensional Fermi surface of a quantum critical fermionic system in d spatial dimensions. For massive fermions, an entropic $c-$function (spatial derivative of the entanglement entropy) can be expressed in the form of differential equations that need to be solved numerically [53–56].

Unlike topologically ordered phases, there is no geometry-independent topological term in the entanglement entropy of free fermionic systems [57, 58]. It is, however, possible to generate such a term even here by placing the system on a multiply connected manifold such as a cylinder, and inserting a magnetic flux through the cylinder [57, 59–62]. It has been shown that if there are ϕ number of electronic flux quanta piercing the system, an additional term is generated in the entanglement entropy of a subsystem of massless Dirac fermions of sufficiently large length (as defined in Ref. [57]) with the form $\frac{1}{6} \ln \left|2 \sin \frac{\phi}{2}\right|$ [57, 60]. Further, there is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [62–64].

The notion of entanglement also plays a key role in the context of holographic duality, which posits that there exists a duality relation between a quantum field theory in $d-$spatial dimensions and a gravity theory in $d+1-$spatial dimensions [65–71]. It is widely believed that the additional dimension can be visualised as a renormalisation group flow that starts from the (almost) critical field theory on the boundary and extends into the bulk [72–81]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a conformal field theory with a theory of quantum gravity through a strong-weak duality relation [82–89]. This has led to new insights in various areas of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [90], the response functions of the Bose-Hubbard model [91, 92], and the phenomenology of non-Fermi liquids [93–96], holographic superconductors [97–102], striped phases [103–105] and holographic Fermi liquids [106–110].

Conversely, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This requires constructing the emergent dimension from first principles by investigating the RG flow of the (often strongly coupled) quantum field theory, and has proved to be considerably more difficult. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory, including [111–120]. These include the momentum-shell renormalisation approach of Refs. [113, 114, 121, 122], the multi-scale entanglement renormalisation ansatz (MERA) approach of Refs. [115–118, 123], the exact holographic mapping (EHM) [124, 125], the Wilsonian RG-based approach of Ki-Seok Kim [126–128] and the unitary renormalisation group (URG) approach [119, 120, 129].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [52, 130] as well as topological features arising from the incompressibility of the Fermi volume [131, 132]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [119, 120, 133]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [134–137]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

Chapter 2

Kondo Breakdown in a Heavy-Fermion Model

2.1 Bilayer extended Hubbard model: Impurity model

[138] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{1\text{p}} + H_f + H_d + H_{fd} \ , \tag{1.1}$$

where $H_{1\text{p}}$ is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{1\text{p}} = -t \sum_{\sigma,\alpha} \sum_{\langle i,j \rangle} \left(c^\dagger_{i,\sigma,\alpha} c_{j,\sigma,\alpha} + \text{h.c.} \right) - \sum_{i,\sigma,\alpha} \mu_\alpha n_{i,\sigma,\alpha} \ , \tag{1.2}$$

such that α sums over the two layers f and d. H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = &-\frac{U_f}{2}(n_{f,\uparrow} - n_{f,\downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_\sigma \left(c^\dagger_{f,\sigma} c_{Z,\sigma,f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,f} c_{Z,\beta,f} \right. \\ &\left. - \frac{W_f}{2} \left(n_{Z,\uparrow,f} - n_{Z,\downarrow,f} \right)^2 \right] \ , \\ H_d = &-\eta_d n_d - \frac{U_d}{2}(n_{d,\uparrow} - n_{d,\downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_\sigma \left(c^\dagger_{d,\sigma} c_{Z,\sigma,d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,d} c_{Z,\beta,d} \right] \ , \end{aligned} \tag{1.3}$$

where $c^\dagger_{\nu,\sigma,\alpha}$ can refer to creation operator for either the $f-$layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_\sigma \left(c^\dagger_{d,\sigma} c_{f,\sigma} + \text{h.c.} \right) \ . \tag{1.4}$$

2.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the "unit" cell of translation. As discussed in the text surrounding eq. **??**, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. Placing the impurity sites at $\boldsymbol{r}_d$ in each layer, we get

$$H_{\text{clust}}(\boldsymbol{r}_d) = H_f(\boldsymbol{r}_d) + H_d(\boldsymbol{r}_d) + H_{fc}(\boldsymbol{r}_d) \ . \tag{2.1}$$

The details of tiling a cluster Hamiltonian were already shown in Sec. **??**; the general idea is to use manybody translation operators (details in Sec. **??**) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_r T^\dagger(\boldsymbol{r}) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}) \\ &= -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.}\right) - \sum_i\left[\eta_d n_{i,d} + \frac{U_d}{2}\left(n_{i,\uparrow,d} - n_{i,\downarrow,d}\right)^2\right] + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,d}\cdot\boldsymbol{S}_{j,d} \\ &\quad - \frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - \frac{U_f - W_f}{2}\sum_i\left(n_{i,\uparrow,f} - n_{i,\downarrow,f}\right)^2 + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,f}\cdot\boldsymbol{S}_{j,f} \\ &\quad - V_\perp\sum_{i,\sigma}\left(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}\right) \end{aligned} \tag{2.2}$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the $d-$layer; a non-zero value moves the layer away from half-filling. The $f-$layer is pinned at half-filling.

Following Sec. **??**, the retarded $k-$space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\boldsymbol{k},\sigma,\alpha;\omega) = \sum_\nu C_{\nu,\boldsymbol{k}\alpha}\mathcal{G}(\mathcal{T}_{\nu,\sigma,\alpha};\omega - \mathcal{E}_\nu(\boldsymbol{k})) \tag{2.3}$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian $H_{1\text{p}}$, with energies $\mathcal{E}_\nu(\boldsymbol{k})$:

$$H_{1\text{p}} = -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.} + f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - V_\perp\sum_{i,\sigma}\left(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}\right) - \sum_i \eta_d n_{i,d} \ , \tag{2.4}$$

and the $\mathcal{T}-$matrices are defined in eq. **??**:

$$\mathcal{T}_{\nu,\sigma,\alpha} = \left[\left(\sum_{\sigma'} c^\dagger_{\boldsymbol{\alpha}\sigma} + \text{h.c.}\right) + \left(S^+_{\boldsymbol{\alpha}} + \text{h.c.}\right) + \left(C^+_{\boldsymbol{\alpha}} + \text{h.c.}\right)\right] c_{\nu,\sigma} \ , \tag{2.5}$$

In order to calculate $\mathcal{E}_\nu(\boldsymbol{k})$, we introduce the $k-$space modes via a Fourier transform:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} c^\dagger_{\boldsymbol{k},\sigma} , \quad f^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} f^\dagger_{\boldsymbol{k},\sigma} . \tag{2.6}$$

In terms of these modes, the $1-$particle Hamiltonian decouples in $k-$space:

$$H_{\text{1p}} = \sum_{\boldsymbol{k},\sigma}\left[(\epsilon_{\boldsymbol{k}} - \eta_d)(n_{\boldsymbol{k},\sigma,d} + n_{\boldsymbol{k},\sigma,f}) - V_\perp(c^\dagger_{\boldsymbol{k},\sigma} f_{\boldsymbol{k},\sigma} + \text{h.c.})\right] . \tag{2.7}$$

The diagonal modes are given by

$$v^\dagger_{\boldsymbol{k},\sigma,\pm} = C_{\pm,k} c^\dagger_{\boldsymbol{k},\sigma} + \sqrt{1 - C^2_{\pm,k}} f^\dagger_{\boldsymbol{k},\sigma} , \tag{2.8}$$

the coefficients obtained from diagonalising the matrix

$$H_{\text{1p}}(\boldsymbol{k}) = \begin{pmatrix} \epsilon_{\boldsymbol{k}} & -V_\perp \\ -V_\perp & \epsilon_{\boldsymbol{k}} - \eta_d \end{pmatrix} . \tag{2.9}$$

The energies are given by $\mathcal{E}_\pm(\boldsymbol{k}) = \epsilon_k - \eta_d/2 \pm \Delta$, with ϵ_k being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$G(\boldsymbol{k}, d; \omega) = C_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},+,d}; \omega - (\epsilon_k - \eta_d/2 + \Delta)) + C_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},-,d}; \omega - (\epsilon_k - \eta_d/2 - \Delta)) . \tag{2.10}$$

Following eq. **??**, we can similarly write down an expression the local lattice Greens function, in real-space:

$$\begin{aligned} &G_{\text{loc}}(\omega) \\ &= \mathcal{G}(d\sigma; \omega) + \sum_{\boldsymbol{k}} \mathcal{G}(d\sigma, \mathcal{T}_{\boldsymbol{k},\sigma}; \omega) \\ &\quad + \sum_{\boldsymbol{k},\alpha=\pm} C_{\alpha,\boldsymbol{k}}\left[\mathcal{G}^>(\mathcal{T}_{\boldsymbol{k},\alpha,d}, d\sigma; \omega - \epsilon_k + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\boldsymbol{k},\alpha,d}, d\sigma; \omega + \epsilon_k - \eta_d/2 - \alpha\Delta)\right] \\ &\quad + \sum_{\boldsymbol{k},\boldsymbol{q},\alpha=\pm} C_{\alpha,\boldsymbol{k}}\left[\mathcal{G}^>(\mathcal{T}_{\boldsymbol{k},\alpha,d}, \mathcal{T}_{\boldsymbol{q},\sigma,d}; \omega - \epsilon_k + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\boldsymbol{k},\alpha,d}, \mathcal{T}_{\boldsymbol{q},\sigma,d}; \omega + \epsilon_k - \eta_d/2 - \alpha\Delta)\right] \end{aligned} \tag{2.11}$$

2.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ESIAM

In the limit of large U_α, we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k},\sigma,\alpha}\epsilon_{\mathbf{k},\alpha} n_{\mathbf{k},\sigma,\alpha} + \sum_\alpha \sum_{Z\in\text{NN}}\left[\frac{1}{2}J_\alpha \sum_{\sigma,\sigma'}\mathbf{S}_\alpha\cdot\boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\sigma,\alpha} c_{Z,\sigma',\alpha} - \frac{W_\alpha}{2}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right] + J\mathbf{S}_f\cdot\mathbf{S}_d . \tag{3.1}$$

The Kondo coupling and bath interaction acquire $k-$dependence:

$$\begin{aligned} J(\mathbf{k},\mathbf{q}) &= \frac{J}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)\right] ,\\ W(\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}') &= \frac{W}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)\right] . \end{aligned} \tag{3.2}$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi,\pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi,-\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q},\sigma,f\rangle$ and $|\mathbf{q},\sigma,d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}},\sigma,f\rangle$ and $|\bar{\mathbf{q}},\sigma,d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_α in the case of $J = 0$:

$$\frac{\Delta J_f(\mathbf{k}_1,\mathbf{k}_2)}{\Delta D} = \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_f(\mathbf{k}_1,\mathbf{q})J_f(\mathbf{q},\mathbf{k}_2) + J_f(\mathbf{q},\bar{\mathbf{q}})W_f(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right] G_f^{(0)}(\mathbf{q}) , \tag{3.3}$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G_f^{(0)}$ is the propagator for the excited intermediate state (for $J = 0$):

$$G_f^{(0)}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2} . \tag{3.4}$$

An identical equations holds for ΔJ_d, with al $f-$quantities replaced with $d-$quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G_f^{(0)} - J\vec{S}_f\cdot\vec{S}_d} . \tag{3.5}$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f\cdot\vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I} . \tag{3.6}$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4}\mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4}\right)\mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} + \frac{1}{1/G_f^{(0)} - J/4}\right)\vec{S}_f\cdot\vec{S}_d . \end{aligned} \tag{3.7}$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α:

$$\frac{\Delta J_\alpha}{\Delta D} = \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha(\mathbf{k}_1, \mathbf{q})J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}})W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] \left(\frac{1}{4}\frac{1}{1/G^{(0)}_\alpha(\mathbf{q}) + 3J/4} \right. \\ \left. + \frac{3}{4}\frac{1}{1/G^{(0)}_\alpha(\mathbf{q}) - J/4}\right) , \tag{3.8}$$

We now turn to the renormalisation of J, arising from hybridisation of $f-$ and $d-$layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_\pm &= \frac{1}{\sqrt{2}}\left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d\right) , \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- , \end{aligned} \tag{3.9}$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α: $|H(q)\rangle_\alpha = c^\dagger_{\mathbf{q},\alpha}c_{\bar{\mathbf{q}},\alpha}|L\rangle_f|L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS^+_fS^-_d$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the $d-$layer followed by a spin-flip of the $f-$layer:

$$\Delta H = \frac{1}{N^2}\sum_{\mathbf{q}\in\text{UV}} \langle L|\frac{1}{2}J_f(\bar{\mathbf{q}}, \mathbf{q})S^+_f c^\dagger_{\mathbf{q}-\downarrow,f}c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H\frac{1}{2}J_d(\mathbf{q}, \bar{\mathbf{q}})S^-_d c^\dagger_{\mathbf{q}\uparrow,d}c_{\bar{\mathbf{q}}\downarrow,d}|L\rangle , \tag{3.10}$$

where $|L\rangle = |L\rangle_f|L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D} , \tag{3.11}$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q},\sigma\in\mathbf{q}_\pm,\alpha}\epsilon_\alpha(\mathbf{q})\tau_{\mathbf{q},\sigma} + \frac{1}{2}\sum_\alpha J_\alpha S^z_\alpha\sum_{\mathbf{q},\sigma}\sigma n_{\mathbf{q},\sigma,\alpha} + J\vec{S}_f\cdot\vec{S}_d . \tag{3.12}$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H = \frac{1}{2}\left(G_f + G_d\right)\left(|H\rangle_+\langle H|_+ + |H\rangle_-\langle H|_-\right) + \frac{1}{2}\left(G_f - G_d\right)\left(|H\rangle_+\langle H|_- + |H\rangle_-\langle H|_+\right) , \tag{3.13}$$

where

$$G_\alpha(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q}) - \frac{1}{4}J} . \tag{3.14}$$

In eq. 3.10, since the first process is an excitation into a $d-$state and the second process is a de-excitation from an $f-$state, the expression can be simplified into

$$\begin{aligned}\langle L|S_f^+ c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S_d^- c^\dagger_{\mathbf{q}+\uparrow,d} c_{\mathbf{q}-\downarrow,d}|L\rangle &= S_f^+ S_d^- \langle H_f|\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H |H_d\rangle \\ &= \frac{1}{2}\left(G_f + G_d\right) S_f^+ S_d^- \ .\end{aligned} \tag{3.15}$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2} \sum_{\mathbf{q}\in \text{UV}} J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) S_f^+ S_d^- \ . \tag{3.16}$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \ . \tag{3.17}$$

The time-reversed of this process can be obtained simply by exchanging the $d-$interaction with the $f-$interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\begin{aligned}\frac{\Delta J}{\Delta D} &= \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \\ &= \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[\frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] ,\end{aligned} \tag{3.18}$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q}, \bar{\mathbf{q}}$.

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger + \mathcal{W}_\alpha \text{Tr}\left[\mathcal{G}'_\alpha\right] \ , \tag{3.19}$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q,q] = \begin{cases} \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2-J/4}\right), & \mathbf{q}\in \text{PS} \ , \\ \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2-J/4}\right), & \mathbf{q}\in \text{HS} \ , \end{cases} \tag{3.20}$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k,k'] = \frac{W}{2}\left[\cos(k_x - k'_x) + \cos(k_y - k'_y)\right]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whole matrix elements are defined as $\mathcal{G}'_\alpha[q,q] = \mathcal{G}[q,q] J_\alpha[q,\bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2}\mathbf{J}\cdot\mathbf{G} \ , \tag{3.21}$$

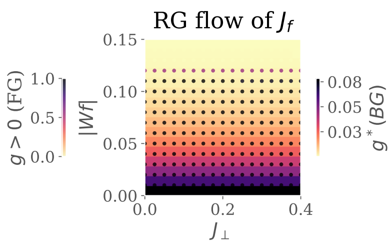

Figure 2.1: RG phase diagrams for various Kondo couplings: $f-$layer (left), $d-$layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the $f-$layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, $1/2$ or 0 respectively.

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q})J_d(\mathbf{q}, \bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q})\left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] . \tag{3.22}$$

2.4 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 2.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the $f-$layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the $d-$layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the $f-$layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the $f-$impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the $f-$layer hybridise with the itinerant electrons of the $d-$layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the $f-$layer forms local moments which decouple from the $d-$layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the $f-$electrons remain itinerant and it is they who hybridise with the $d-$electrons to again form two bands separated by a hybridisation gap.

2.5 Results at half-filling

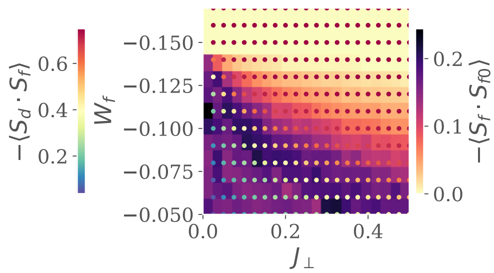

Figure 2.2: *Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the $f-$impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the $f-$impurity and the $d-$impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:*

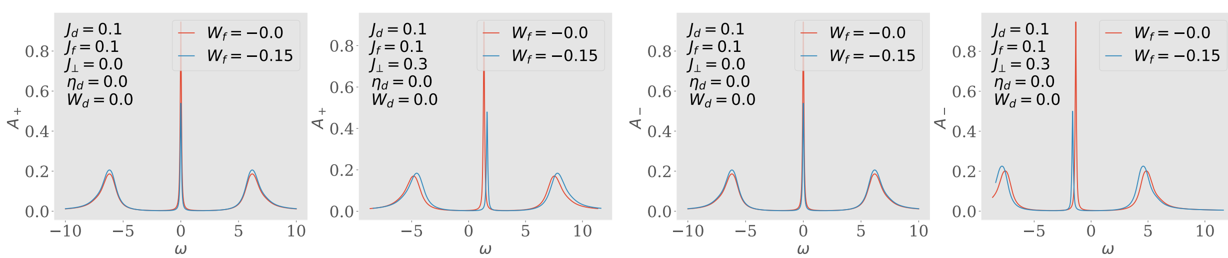

Figure 2.3: *Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.*

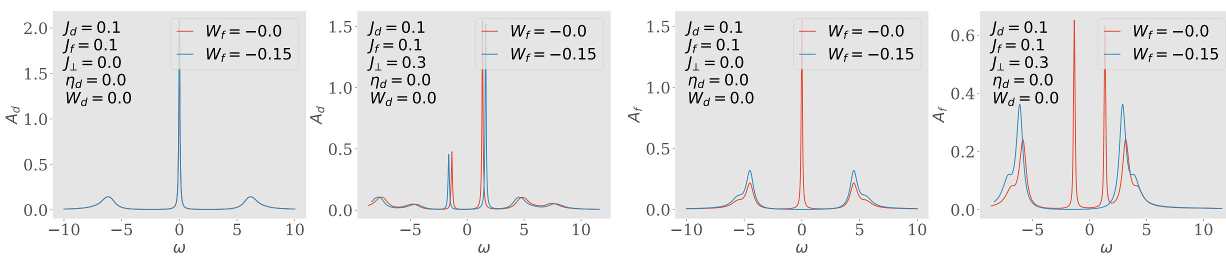

Figure 2.4: *Spectral functions A_d and A_f of the auxiliary model, for excitations in the $d-$ and $f-$layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f.*

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